NASA Space Apps Challenge 2026

Earth Information Jukebox

Hear the Arctic Ice Disappear

NASA-based satellite data turned into sound

The problem

Charts are hard to feel

Satellites have tracked Arctic sea ice since 1979, but the record usually appears as lines on a graph.

Charts are not for everyone

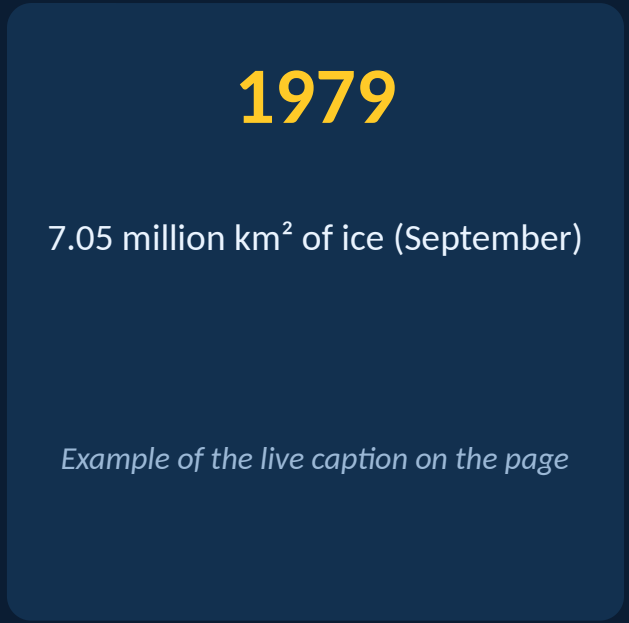
People who are blind or have low vision cannot easily explore data shown only visually.

Climate change feels distant

Long-term trends are easy to ignore when they are just numbers.

Our solution: a jukebox for Earth data

- Each year is one musical note.
- Low note = a lot of sea ice. High note = less ice.
- A chart marker and live caption show the year and ice extent as each note plays.
- One Play button. No install, works in any browser.



How it works

1

NSIDC Sea Ice Index

Monthly Arctic extent, from NASA satellite observations

2

Python + Pandas

Cleaned the data in Google Colab, kept the September value per year

3

Data to pitch

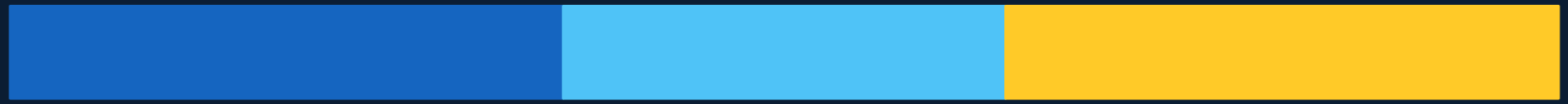
More ice = 220 Hz, less ice = 880 Hz

4

Web page

HTML, JavaScript and Web Audio API, hosted on GitHub Pages

What you hear



LOW NOTE · lots of ice

220 Hz

HIGH NOTE · less ice

880 Hz

Over more than 45 years of data, the pitch climbs as the Arctic ice shrinks. Open the live demo to listen.

Data, links and next steps

Data: NSIDC Sea Ice Index (NOAA at NSIDC), based on NASA satellite observations

Live demo: https://joselynrajeev55-ops.github.io/earth-information-jukebox/

Code: https://github.com/joselynrajeev55-ops/earth-information-jukebox

Next: more tracks (Antarctic ice, sea surface temperature), satellite imagery, user controls